

The project was developed using a set of powerful Python-based tools and libraries for data processing, machine learning, visualization, and deployment. These technologies provide the foundation for efficient model development, analysis, and application deployment.

**Anaconda Navigator:** A desktop graphical user interface used to launch applications and manage Python packages and environments for data science. <https://www.anaconda.com/download>

**PyCharm:** An Integrated Development Environment (IDE) specifically designed for Python programming, offering code analysis and debugging tools. <https://www.jetbrains.com/pycharm/>

**NumPy:** Core library for numerical computing in Python, providing support for large, multi-dimensional arrays and mathematical functions. <https://numpy.org/doc/stable/>

**Pandas:** Data manipulation and analysis library offering data structures and operations for manipulating numerical tables and time series. <https://pandas.pydata.org/docs/>

**Scikit-learn:** Machine learning library featuring classification, regression, and clustering algorithms like support vector machines and random forests. <https://scikit-learn.org/stable/>

**Matplotlib:** Comprehensive library for creating static, animated, and interactive visualizations in Python. <https://matplotlib.org/stable/>

**Seaborn:** Data visualization library based on matplotlib that provides a high-level interface for drawing attractive statistical graphics. <https://seaborn.pydata.org/>

**Flask:** A lightweight WSGI web application framework used to build web APIs and deploy machine learning models. <https://flask.palletsprojects.com/>